LIGHTYEAR

MS83 Native iDempiere context and migration measurement

Delivered by PR #214, merged 26 September 2026 as `45288de`.

Status: native measurement completed; differences and validity findings retained. The follow-up MS84 resolves these findings on a documented, corrected execution path. It does not replace this original result.

Customer benefit

Before promising that a migration works, establish what the tools can actually prove and what the databases actually do. This milestone replaces assumptions about iDempiere's database structure and migration effects with observations from running Oracle and PostgreSQL databases.

In beginner terms, a migration script is a set of instructions that changes a database. A successful exit says the instructions ran. It does not say that both databases ended with equivalent data, or that all their objects remain valid. MS83 measures those questions separately.

Delivered behavior

The experiment imported the official release-13 schema and data into local Oracle AI Database 26ai Free (23.26.3.0.0) and PostgreSQL 15.19. Source, seed archives and database images are pinned in the retained evidence. Both engines' catalogs and effective session settings were captured through versioned queries. Capture fails if a required query fails; completeness refers to those schema-owner queries, not every database-wide setting.

Stage one supplied those observed facts to the static comparator and reran all 1,078 pinned script pairs against a same-version control without the facts. Stage two executed the 20 normal migration pairs outstanding on both seeds, capturing complete ordinary base-table rows and defined structural metadata before and after each pair. Raw UUIDs, timestamps and registration paths were retained rather than discarded to obtain a passing result.

Measurement	Result at PR #214
Static script pairs replayed	1,078
SQL units decided without captured context	780 / 111,293
SQL units decided with captured context	780 / 111,293
Additional decisions from captured context	**0**
Pending migration pairs executed natively	20
Successful native client executions	40 / 40
Strict whole-pair row-effect comparisons that differ	20 / 20
Exact changed-table delta matches	49 / 111
Further table comparisons with matching changed-field transitions	13
Invalid Oracle objects after the tranche	14
Whole-pair equivalence claims	**0**

The earlier improvement from 776 to 780 decided units belongs to MS81. MS83 does not count it again. Zero additional decisions establishes the limit reached by this gate on this corpus with these supplied facts; it is not a universal limit on static analysis.

Findings and why they matter

The official seeds already differed: Oracle started with 913 base tables and 185,292 rows; PostgreSQL had 912 tables and 185,253 rows. Some final differences could therefore predate execution. The 13 matching changed-field comparisons were useful diagnostics, but did not make the strict whole-pair verdict pass.

All 20 scripts registered on both engines, yet registration data, sequence counters, clocks and application metadata still differed. Fourteen Oracle views or procedures were invalid after the tranche. These observations showed why successful script execution alone was an insufficient acceptance test.

A hypothetical evaluator might see two scripts exit successfully and conclude that an upgrade is safe. This milestone instead gives them the data differences and object-validity findings needed to withhold that conclusion and commission a specific repair and replay plan.

Evidence and limits

The runtime report and sealed receipt bind 123 published evidence files: native captures, import and execution logs, comparisons, diagnostics, final checks and the controllers used for the experiment. Hashes establish integrity, not independent attestation. Full row snapshots and private connection settings remain outside Git.

Of the 1,078 inventory pairs, 1,053 were already registered on both seeds and were not freshly executed. One older Oracle-only registration gap and four maintenance pairs were also outside the executed tranche. The release-13 seed cannot recreate historical entry states simply by replaying old scripts over it.

No application equivalence, historical-upgrade certification, Oracle behavior- catalog conformance or platform qualification is claimed. Execution used local Docker; no GCP resources were started. Containers were stopped with data retained.

Validation and relationship to MS84

Collector, comparator, context-binding and publication-integrity tests ran with the calibration checks on Windows and Linux. CI also reproduced the static measurement against the pinned public source. Native logs and validity checks provide the execution evidence; synthetic tests do not add native coverage.

MS83 established the questions. MS84 diagnoses registration, creates comparable starting data, validates explicit rules for expected runtime differences, resolves validity findings, and replays history from an older seed.